LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Linear analysis, MATB34

Written Assignment

Deadline: October 13, 2025, 13.00

Solve the following problems in written (or handwritten) and submit before the deadline in canvas.

1. Compute the Laplace transform for the following functions

a) $\cos^2(\omega t)$;

b) $t^2 - 1$;

c) $e^{-t}\cos^2 t$.

2. Find the inverse Laplace transform for the functions

a) $\frac{1}{(s-a)(s-b)}$;

b) $\frac{s^2}{1+s^2}$.

3. Using the Laplace transform find particular solutions to the differential equations

a) $y'' - y' + 2y = 2e^{3t}, y(0) = y'(0) = 0$;

b) $y'' + 4y' + 4y = t^3 * e^{-2t}, y(0) = 1, y'(0) = 2$;

c) $y''' + 3y'' + 3y' + y = 6e^{-t}, y(0) = y'(0) = y''(0) = 0$;

d) $y'''' + 2y'' + y = \sin(t), y(0) = y'(0) = y''(0) = y'''(0) = 0$.